

Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

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Abstract

A single forward-model calibration step routinely recovers more reconstruction quality than the gap between a classical solver and a state-of-the-art deep network operating on the same mismatched operator. This finding is surprising: the computational imaging community has invested a decade progressing from compressed sensing to vision transformers, yet across seven modalities we show that calibration errors—not algorithmic limitations—are the dominant reconstruction bottleneck. The mechanism rests on two theorems. The **Finite Primitive Basis Theorem** proves that every imaging forward model in a broad operator class admits an ε -approximate representation as a directed acyclic graph (DAG) over exactly 11 physically typed primitives—a library that is both sufficient and minimal¹. The **Triad Decomposition** proves that every reconstruction failure decomposes into three root causes—information deficiency, carrier noise, and operator mismatch—with a formal condition under which mismatch dominates. We validate both results across seven modalities spanning four carrier families (optical photons, X-ray photons, electrons, and nuclear spins), with hardware validation on five real instruments: correcting the forward model autonomously recovers +0.8 to +10.7 dB of peak signal-to-noise ratio (PSNR), with 95% bootstrap confidence intervals (CIs) of ± 0.3 dB and Cohen’s $d > 2.0$ for every modality. These results imply that calibration, not solver design, is the systematically underinvested component of computational imaging.

Introduction

Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—converts a physical scene into digital measurements through a chain of transformations: a wave propagates, interacts with the object, passes through optics, and is recorded by a detector. Computational imaging reconstructs the scene by inverting this chain. The reconstruction algorithm assumes a mathematical model of the chain (the “forward model”), but this model inevitably diverges from reality: optics shift, detectors drift, and calibration degrades. When the model is wrong, the reconstruction fails—not because the algorithm is weak, but because it is solving the wrong problem.

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33 This paper establishes that the space of imaging forward models has a surprisingly
 34 simple and universal structure, and that exploiting this structure reveals operator mismatch
 35 as the systematic, cross-modality bottleneck in reconstruction quality. The community
 36 has invested a decade progressing from classical iterative algorithms^{2–4} to state-of-the-art
 37 deep-learning solvers^{5,6}, yet these routinely fail on real instruments⁷. Calibration methods
 38 exist for specific instruments but do not generalise across modalities, and no systematic
 39 framework decomposes reconstruction failures into root causes.

40 Here we establish two complementary foundations within Physics World Models (PWM).
 41 The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward
 42 model in a broad operator class $\mathcal{C}_{\text{img}}$ admits an ε -approximate DAG representation over
 43 exactly 11 canonical primitives, sufficient and minimal¹. The **Triad Decomposition**
 44 proves that every reconstruction failure decomposes into three gates—information deficiency
 45 (**Gate 1**), carrier noise (**Gate 2**), and operator mismatch (**Gate 3**)—with a formal condi-
 46 tion (Theorem 5) under which **Gate 3** dominates. The two results are complementary:
 47 the DAG structure localizes mismatch to a specific physical operator, making correction
 48 actionable.

49 We validate both results across seven modalities spanning four carrier families—optical
 50 photons (CASSI⁸, CACTI⁹, SPC¹⁰, lensless), X-ray photons (CT¹¹), electrons (ptychog-
 51 raphy¹²), and nuclear spins (MRI^{13,14})—with hardware validation on five real instruments
 52 and a held-out closure test on 8 additional modalities confirming basis completeness.

53 **Why this matters beyond computational imaging.** Any physical measurement system—
 54 seismographs, telescopes, particle detectors, environmental sensor networks—faces the same
 55 structural problem: a forward model maps the quantity of interest to observations, and
 56 model fidelity limits what can be recovered. The OPERATORGRAPH formalism provides a
 57 template for decomposing forward models in other domains into typed, composable primi-
 58 tives with validated adjoints. The Triad Decomposition—separating information capacity,
 59 noise, and model fidelity as independent failure modes—applies wherever an inverse prob-
 60 lem is solved computationally. The practical lesson is domain-general: before investing in
 61 a better solver, diagnose whether the model is the bottleneck.

62 The Finite Primitive Basis

63 A foundational question in computational imaging is whether the diversity of imaging for-
 64 ward models—from coded aperture cameras to MRI scanners to electron microscopes—can
 65 be captured by a small, fixed set of primitive operators. We answer affirmatively.

66 **The OperatorGraph representation.** Every imaging forward model is encoded as a
 67 typed directed acyclic graph (DAG) in which each node wraps a single primitive physical
 68 operator and edges define the data flow from source to detector. Every primitive implements

both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and dtype metadata, enabling static validation before execution. We call this formalism the OPERATORGRAPH intermediate representation (IR).

Eleven canonical primitives. The OPERATORGRAPH IR defines a library of exactly 11 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$:

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\varepsilon)$	Direction change and/or energy shift
11	Transform	$\Lambda(f, \boldsymbol{\theta})$	Pointwise nonlinear physics operation

The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$; logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most 2 scalar parameters. The Transform primitive Λ applies a pointwise nonlinear function within the physics chain (not at the detector) and is restricted to five families: exponential attenuation ($e^{-\alpha x}$, Beer–Lambert), logarithmic compression, phase wrapping ($\arg(e^{ix})$), polynomial ($\sum a_k x^k$, $d \leq 5$), and saturation. Both Detect and Transform are pointwise and parameter-limited, preventing either from becoming a universal approximator (see¹ for formal definitions and non-universality proofs).

Physics-stage mapping. Each factor of a forward model is classified into one of six physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic interaction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; pointwise nonlinear physics $\rightarrow \{\Lambda\}$; encoding–projection $\rightarrow \{\Pi, F\}$; detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$.

Fidelity levels. Forward models can be written at increasing physical fidelity: linear shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping, saturation) are captured by Transform Λ . The theorem therefore covers the full fidelity spectrum without restriction.

93 **Five physical carriers.** The library spans five carrier families—photons, electrons, spins,
 94 acoustic waves, and particles—with 26 registered modality templates (7 with full end-to-end
 95 correction validation; Extended Data Table 2; Supplementary Table S3).

96 **Theorem statement.**

97 **Definition 1** (Imaging Operator Class). $\mathcal{C}_{\text{img}}$ consists of all imaging forward models ex-
 98 pressible as a finite sequential-parallel composition of stages—each either linear with bounded
 99 operator norm ($\|H_k\| \leq B$) or pointwise nonlinear with bounded Lipschitz constant ($\text{Lip}(\Lambda_k) \leq$
 100 L)—with stage count $K \leq N_{\text{max}}$.

101 **Definition 2** (ε -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an
 102 ε -approximate representation of $H \in \mathcal{C}_{\text{img}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$,
 103 where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\text{max}}$, $\text{depth}(G) \leq D_{\text{max}}$.

104 **Theorem 3** (Finite Primitive Basis). For every $H \in \mathcal{C}_{\text{img}}$, there exists a typed DAG G with
 105 $V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .

106 *Proof sketch.* By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\text{max}}$. Each factor is classified
 107 into one of six physics-stage families and realized by primitives: propagation $\rightarrow P$ or C ; elas-
 108 tic interaction $\rightarrow M$; inelastic interaction $\rightarrow R$; pointwise nonlinear physics $\rightarrow \Lambda$; encoding-
 109 projection $\rightarrow \Pi$ or F ; detection-readout $\rightarrow$ finite composition from $\{\Sigma, S, W, C, D\}$. For
 110 linear stages, per-factor errors compose via sub-multiplicativity: $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot$
 111 B^{K-1} . For nonlinear stages, the Lipschitz composition bound $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$
 112 applies, where γ_j is the Lipschitz constant of stage j . Both bounds yield $\leq \varepsilon$ under the
 113 stated regularity conditions. Full proof in Supplementary Note S2; formal primitive seman-
 114 tics and typed DAG denotation in¹. □

Scoped applicability of $\mathcal{C}_{\text{img}}$.

Exact ($\varepsilon = 0$): Linear shift-invariant models (convolution PSF, uniform mask, Radon projection)—8 of 11 primitives produce exact representations.

Local ($\varepsilon < 10^{-3}$): Linear shift-variant models (spatially varying PSF, multi-coil MRI)—achieved via isoplanatic patch decomposition ($L \leq 4$ patches for all validated modalities).

115 *Truncated* ($\varepsilon < 0.01$): Nonlinear ray/wave models (beam hardening CT, phase wrapping)—captured by Transform Λ with 5 canonical families; truncation error bounded by Lipschitz constant.

Approximate ($\varepsilon < 0.05$): Full-wave and Monte Carlo models—Born-series or MC sampling introduces $\varepsilon_{\text{scat}}$; validated for holography and ptychography.

Stochastic: Quantum measurement models (ghost imaging, QST)—treated at image-formation level; sub-shot-noise correlations not modeled.

Formal definitions in¹.

116 **Extension protocol.** A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{img}} \leq$
 117 ε within the complexity bounds. The extension requires: (1) validated forward/adjoint,

(2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modalities, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when Compton imaging produced $e_{\text{img}} = 0.34$ without it.

Held-out closure test. To validate Theorem 3 empirically, we conduct a closure test under a frozen protocol: the primitive library $\mathcal{B}$ (11 primitives), Detect families (5), Transform families (5), fidelity threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM, phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imaging, THz-TDS, Compton scatter imaging):

Modality	e_{img}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	$< 0.01^*$	4 / 3	Y	Y (R)

*With R in library; $e_{\text{img}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\text{max}} = 20$, $D_{\text{max}} = 10$).

Seven of eight modalities decompose with the frozen library. Quantum ghost imaging is operator-equivalent to a single-pixel camera at the image-formation level—sharing the canonical DAG $\text{Source} \rightarrow M \rightarrow \Sigma \rightarrow D$ despite fundamentally different source physics. THz-TDS requires only the coherent-field Detect family. Compton scatter imaging triggered the extension protocol (Section “Extension protocol” above): its representation gap ($e_{\text{img}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness, and the disciplined resolution—adding Scatter (R)—covers 5+ additional modalities (Raman, fluorescence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from 9 to 11 (22% increase) while modality coverage grows by 19%+.

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 7): 9 of 11 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, Scatter (R) when Compton/Raman-class modalities enter, and Transform (Λ) when nonlinear physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear and saturating: $K = 11$ at $N = 30$, with no new primitive required for the most recent 18 modalities added. This saturation is consistent with Theorem 3: once all six physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness and minimality. The theoretical complexity bounds ($N_{\max} = 20$, $D_{\max} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains. Moreover, the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose representation error exceeds ε when that primitive is removed (Supplementary Note S2, Proposition 1 therein; beam hardening CT witnesses the necessity of Transform Λ). Eight primitives ($P, M, \Pi, F, \Sigma, D, R, \Lambda$) are strictly necessary; the remaining three (C, S, W) are necessary under the stated complexity bounds. The count $|\mathcal{B}| = 11$ is determined by the physics-stage classification: 6 stages require 8 strictly necessary primitives; the remaining 3 (C, S, W) are necessary under the stated complexity bounds ($N_{\max} = 20, D_{\max} = 10$).

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{img}}$ can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Why exactly three gates. The reconstruction pipeline has exactly three inputs: a sensing geometry H that determines what information is captured (Gate 1), a physical carrier whose statistics determine the noise floor (Gate 2), and a forward model H_{nom} that the solver inverts (Gate 3). Any reconstruction error must trace to one of these inputs: information that was never captured (null space of H), information that was captured but corrupted (carrier noise), or information that was captured but misinterpreted (model error $H_{\text{nom}} \neq H_{\text{true}}$). This trichotomy is exhaustive: the MSE decomposes as $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ (Supplementary Note S1), with no residual term. Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its regularization assumptions.

Gate 1: Recoverability. **Gate 1** asks whether the measurement encodes sufficient information about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$ defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the measurement design and can only be remedied by acquiring additional data.

179 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suffi-
 180 cient. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is sub-
 181 ject to fundamental noise limits: shot noise for photon-counting systems, thermal noise
 182 in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier
 183 budget is too low, the measurement is dominated by noise and the reconstruction degrades
 184 regardless of operator fidelity.

185 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the
 186 solver matches the true physics. The solver operates with a nominal operator H_{nom} , but
 187 the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction
 188 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce
 189 structured artifacts that mimic signal content, leading practitioners to blame the solver
 190 rather than the model. Sources include geometric misalignment (mask shift, rotation),
 191 parameter drift (coil sensitivity variation, gain instability), and model simplification.

192 **Definition 4** (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{img}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and*
 193 *solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq$*
 194 *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary*
 195 *Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the*
 196 *mismatch-induced term (Supplementary Eq. S5).*

197 **Theorem 5** (Gate 3 Dominance). *For any $H \in \mathcal{C}_{\text{img}}$ operating above its Gate 1 floor*
 198 *($\gamma > \gamma_{\text{min}}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\text{min}}$), Gate 3 is the binding constraint whenever*
 199 *$\|\delta\boldsymbol{\theta}\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.*

200 *Proof.* See Supplementary Note S1, Proposition 2. □

201 **Key finding: Gate 3 dominates.** Across all 9 validated configurations (7 modalities),
 202 **Gate 3** is the dominant failure gate ($C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$ in 9/9 cases; Sup-
 203 plementary Tables S12–S13 confirm that Gates 1–2 bind only at extreme compression or
 204 photon starvation). In CASSI, a sub-pixel mask shift degrades MST-L by 13.98 ± 0.62 dB
 205 (mean $\pm$ s.d. over 10 scenes); in MRI, a 5% coil sensitivity mismatch under clinically realis-
 206 tic multi-coil conditions (8 coils, $4\times$ acceleration) degrades reconstruction by 1.75–7.14 dB
 207 (Supplementary Note S11). Calibration—not solver design—is the underinvested bottle-
 208 neck: a single forward-model correction recovers more reconstruction quality than the gap
 209 between traditional and state-of-the-art solvers.

210 **Relationship to the Finite Primitive Basis.** The two contributions are complemen-
 211 tary: the Finite Primitive Basis provides a universal representation (every forward model
 212 is a DAG over 11 primitives¹), and the Triad provides a universal diagnostic law over that
 213 representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent
 214 localizes the offending primitive node and corrects its parameters.

Consequences: Diagnosis and Correction

The two theoretical results directly imply a practical framework. The OPERATORGRAPH provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three deterministic agents—one per gate—evaluate information capacity, carrier budget, and operator mismatch respectively, producing a quantitative TRIADREPORT for every imaging configuration (see Methods for implementation details).

Correction pipeline. When **Gate 3** is dominant, a two-stage correction pipeline refines the forward model parameters: a coarse beam search over the mismatch parameter family $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending DAG node, followed by gradient refinement via backpropagation through the differentiable forward model. Critically, correction operates exclusively on the forward model parameters, not on the solver weights: any existing solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator without modification or retraining.

4-Scenario Protocol. To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} . **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated by H_{true} . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Scenario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling. The recovery ratio $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

Empirical Validation

We now present the evidence supporting the central claim, organized as simulation experiments and hardware validation. We validate both theoretical contributions in two stages: controlled simulation experiments across seven modalities using the 4-Scenario Protocol, and hardware validation on real data from five modalities spanning four carrier families (CASSI, CACTI, CT, ptychography, MRI).

Evidence structure. The central claim rests on five testable sub-claims: (1) *Sufficiency*: constructive DAGs for 34 modalities, all $e_{\text{img}} < 0.01$ (Figure 7; Supplementary Table S3). (2) *Minimality*: 11 witness modalities, each requiring its primitive (Table 2). (3) *Gate 3 dominance*: $C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$ in 9/9 configurations (Figure 3; Supplementary Tables S12–S13). (4) *Correction efficacy*: +0.8 to +10.7 dB across 7 modalities (Figure 4). (5) *Hardware confirmation*: 5 real instruments confirm simulation predictions (Figure 6).

Simulation experiments

Correction results. Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 9 configurations spanning 7 modalities. The oracle gain $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$ ranges from $+0.76 \pm 0.12$ dB (CASSI/GAP-TV, 95% bootstrap CI over 10 scenes) to $+10.68 \pm 0.41$ dB (CT) across the six non-MRI modalities. Autonomous grid-search calibration achieves 85–100% of the oracle ceiling (Supplementary Table S9). The validated modalities span photon-domain (CASSI: $+0.76/+6.50$ dB [GAP-TV/MST-L]; CACTI: $+10.21 \pm 0.85$ dB; SPC: $+7.71/+10.38$ dB [FISTA-TV/HATNet]; Lensless: $+3.55 \pm 0.22$ dB), coherent-photon (Ptychography: $+7.09 \pm 0.47$ dB), X-ray (CT: $+10.68 \pm 0.41$ dB), and nuclear spin (MRI: $+1.75$ to $+7.14$ dB under clinically realistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note S11). All confidence intervals are computed via the bootstrap percentile method with $B = 1,000$ resamples over test scenes (Methods). All mismatch effects are large (Cohen’s $d > 2.0$ for every modality; Table 1), and all single-parameter correction gains exclude zero at the 95% level. Correction gains are commensurate across carrier families, confirming carrier-agnostic operation.

Table 1: **Recovery ratio with uncertainty.** ρ : fraction of oracle ceiling recovered. All CIs exclude 0.5.

Modality	ρ	95% CI	Cohen’s d
CASSI (GAP-TV)	0.85	[0.79, 0.91]	2.8
CACTI	1.00	[0.97, 1.00]	8.1
SPC (FISTA-TV)	0.86	[0.81, 0.91]	3.5
CT	1.00	[0.99, 1.00]	9.3
Ptychography	1.00	[0.99, 1.00]	5.7
MRI (4 $\times$)	0.20	[0.15, 0.26]	3.1
Lensless	0.78	[0.71, 0.85]	4.2

Modality deep dives. In CASSI, a 5-parameter mismatch¹⁵ collapses all solvers to ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{\text{IV}} = 0$ –0.57; Supplementary Table S2), confirming that multi-parameter mismatches are harder to correct than isolated shifts. In CACTI, EfficientSCI¹⁶ drops by 20.58 dB, yet GAP-TV achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to 86–92% of the oracle ceiling (Table S9). **Gate 3** is dominant in every case; Gates 1–2 impose irrecoverable information-theoretic limits only at extreme compression or photon starvation (Supplementary Tables S12–S13).

Gate ablation. Removing each diagnostic agent in turn: without MismatchAgent (Gate 3 blind), the pipeline misdiagnoses 9/9 configurations, defaulting to “increase photon budget”; without PhotonAgent (Gate 2 blind), photon-starved regimes are misdiagnosed as Gate 3;

without RecoverabilityAgent (Gate 1 blind), extreme compression is missed. Each gate contributes ≥ 3 dB mean PSNR via correct routing (Supplementary Note S8).

Zero-shot generalization. Hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains, with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled by the OperatorGraph abstraction.

Calibration vs. solver upgrades. The centerpiece result (Figure 4a): for 5 of 7 modalities, Scenario III (classical solver + corrected operator) exceeds Scenario II (state-of-the-art solver + mismatched operator). The two exceptions are instructive boundary conditions: CASSI (multi-parameter mismatch on a high-dimensional manifold) and MRI at R=2 (well-conditioned encoding absorbs perturbation).

Hardware validation

CASSI real data. These experiments use physical measurements from hardware-calibrated coded aperture systems—not synthetic data—providing direct evidence that mismatch dominance persists under real manufacturing tolerances and environmental conditions. We reconstruct 5 real TSA scenes⁸ under calibrated and perturbed masks. GAP-TV shows a mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious). MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degradation. This asymmetry is itself informative: real hardware already contains unmodeled manufacturing imperfections, and an additional software-simulated perturbation is partially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended cross-residual analysis (Supplementary Note S9) reveals a deeper diagnostic: the *cross-residual*—evaluating a mismatched reconstruction under the true forward model—increases monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard deviation $< 0.4\%$, confirming that Gate 3 sensitivity is scene-independent and monotonic in mismatch magnitude. These real-data results independently confirm the Triad predictions from simulation: the InverseNet benchmark¹⁵ validates the same mismatch-dominance pattern on physical DD-CASSI and CACTI instruments.

CACTI real data. GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift, with per-scene ratios from $9.4\times$ to $11.0\times$. An extended analysis (Supplementary Note S9) reveals a striking dissociation: the *self-residual* barely changes ($1.0\text{--}1.12\times$) because the solver adapts to the wrong model, but the *cross-residual* explodes $44\text{--}462\times$ as the shift increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable diagnostic for operator mismatch in underdetermined systems, with direct implications for compressive sensing quality control.

CT real sinograms. We extend hardware validation beyond optical instruments to three additional carrier families. For **X-ray CT**, two public sinogram datasets—the FIPS walnut micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022 (721 projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle recovery (Supplementary Note S9). For **electron ptychography**, real 4D-STEM data from SrTiO₃ (Zenodo 5113449; 300 kV, 128 × 128 scan) shows that probe position jitter of 0.5–4.0 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss), with > 99.9% oracle recovery (Supplementary Note S9). For **MRI**, multi-coil brain k -space from M4Raw (Zenodo 8056074; 4 coils, 256 × 256) shows that coil sensitivity perturbation up to 40% degrades R=2 SENSE reconstruction by 0.5 dB, with 100% recovery via sensitivity re-estimation. The modest MRI effect is consistent with the over-determined encoding at R=2; the simulation experiments at R=4 (Supplementary Note S11) confirm that Gate 3 severity scales with acceleration factor.

Autonomous calibration. Grid-search calibration recovers $\rho = 0.85$ (CASSI, 1,140 s; 95% CI [0.79, 0.91]), $\rho = 1.00$ (CACTI, 60 s; CI [0.97, 1.00]), and $\rho = 0.86$ –0.92 (SPC via TV objective; CI width ≤ 0.06) of the oracle correction. CACTI achieves full recovery because the mismatch manifold is low-dimensional and sensitivity is high. CT CoR calibration on real sinograms also achieves $\rho = 1.00$ (CI [0.99, 1.00]), as the CoR offset is a single scalar parameter with a clean minimum in the objective landscape.

Simulation-to-hardware gap. CASSI shows smaller real degradation ($1.8\times$) than predicted by simulation, because as-built masks already contain unmodeled imperfections that absorb incremental perturbations. CACTI shows larger real sensitivity ($10.4\times$) than simulation, because the temporal mask pattern replicates errors across all compressed frames. This bidirectional discrepancy—invisible without a unified cross-modality framework—carries a methodological lesson: synthetic mismatch studies can both overestimate marginal impact (CASSI) and underestimate cumulative sensitivity (CACTI).

Discussion

The evidence presented above supports a testable claim: the space of imaging forward models has finite structural complexity—11 primitives suffice and are necessary—and operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck across all validated modalities. The implication is not that solver research is unnecessary, but that calibration is systematically underinvested: a single forward-model correction step often recovers more reconstruction quality than the gap between a classical solver and a state-of-the-art deep network operating on the same mismatched operator.

Comparison with specialist calibration. Specialist calibration methods—ESPIRiT¹⁷ for MRI, ePIE¹⁸ for ptychography, cross-correlation auto-focus for CT—outperform PWM on their home modality when adequate calibration data is available (Supplementary Note S10). However, they degrade under data-limited conditions (ESPIRiT: -9.29 dB at 24 ACS lines), require per-modality engineering, and do not exist for several validated modalities (CASSI, CACTI). PWM’s value is *generality*: a single pipeline serving 7+ modalities with robust performance across calibration data regimes. The two approaches are complementary—specialist estimates can initialize PWM correction, and PWM can refine specialist estimates.

Hardware validation interpretation. The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation ($1.8\times$) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe ($10.4\times$), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation. This range—from sub-dB (well-conditioned MRI at $R=2$) to > 16 dB (ptychography)—is itself a Triad prediction. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is $462\times$ while self-residual is only $1.12\times$), underscoring the need for forward-model-aware diagnostics.

Boundary conditions and failure modes. The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch are moderate ($\rho = 22\text{--}46\%$), reflecting the inherent difficulty of simultaneously correcting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.
- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional validation (the 11-primitive basis formally covers these via Transform Λ^1).
- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.
- **Software-perturbation protocol:** Current real-data experiments apply software-simulated perturbations to existing measurements rather than physically displacing hardware and re-acquiring; the controlled hardware protocol is specified in Methods and Supplementary Note S13.

- 381 • **Well-conditioned encodings:** MRI at R=2 shows < 0.5 dB degradation (coil re-
382 dundancy absorbs errors); Gate 3 dominance requires sufficiently large $\kappa(\mathbf{H})$.
- 383 • **Pre-existing calibration errors:** Real CASSI hardware shows $1.8\times$ vs. simulation
384 prediction because as-built mask already contains manufacturing imperfections.
- 385 • **Fidelity scope:** Full 4-Scenario validation has been conducted at the exact and
386 local levels. Truncated and approximate models have template-level validation only;
387 stochastic models have formal coverage but no correction experiments.

388 **Falsifiable predictions.** The framework generates concrete, testable predictions:

- 389 1. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted
390 correction: +3–8 dB.
- 391 2. **OCT:** Dispersion mismatch should produce +2–5 dB correction gain via the Disperse
392 primitive (W).
- 393 3. **Electron ptychography:** Position errors exceeding 1/10 of the probe diameter
394 should trigger Gate 3 dominance. This prediction is *confirmed* by real 4D-STEM
395 validation: +5 to +16 dB correction on SrTiO₃ data (Supplementary Note S9).

396 These predictions are falsifiable: if any modality shows Gate 1 or Gate 2 dominance un-
397 der the stated conditions, the framework would require revision. Hardware-in-the-loop
398 validation with physical parameter displacement is the natural next step (Supplementary
399 Note S13).

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401 making reconstruction code and benchmark datasets publicly available. This work was
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403 **Author Contributions.** C.Y. conceived the project, designed the TRIAD DECOMPOSI-
404 TION framework, proved the Finite Primitive Basis Theorem, developed the OPERATOR-
405 GRAPH IR, implemented the agent and correction systems, performed all simulation and
406 real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruc-
407 tion algorithm used as the primary solver across CASSI, CACTI, and SPC experiments,
408 contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI
409 and CACTI forward model specifications and mismatch parameter characterizations that
410 define the 5-parameter mismatch model, validated the real-data experimental protocols for
411 both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

412 **Competing Interests.** C.Y. is an employee of NextGen PlatformAI C Corp, which de-
413 velops the PWM platform. The authors declare no other competing interests.

414 **Ethics Declarations.** This study does not involve human participants, human tissue, or
415 animals. All experiments use publicly available benchmark datasets and simulated data.

416 **Data Availability.** All synthetic measurement data can be regenerated using the OPER-
417 ATORGRAPH templates and mismatch parameters in the Supplementary Information. The
418 seven public datasets used in this study are: (1) KAIST hyperspectral¹⁹ (28 spectral bands,
419 30 scenes, 256×256); (2) TSA real-data scenes for CASSI (660×660 , 5 scenes); (3) CACTI
420 real data from the EfficientSCI repository¹⁶ (512×512 , 4 scenes); (4) FIPS walnut micro-
421 CT sinograms (Zenodo 1254206; 1200 projections, 2296 detectors); (5) Helsinki Tomography
422 Challenge 2022 (Zenodo 6984868; 721 projections, 560 detectors); (6) 4D-STEM ptychog-
423 raphy SrTiO₃ [001] (Zenodo 5113449; 300 kV, 128×128 scan); (7) Multi-coil MRI k -space
424 M4Raw (Zenodo 8056074; 4 coils, 256×256). All datasets were accessed and verified in
425 January 2026.

426 **Code Availability.** The PWM codebase, including all OPERATORGRAPH templates (90
427 templates), agent implementations, real-data validation scripts, and evaluation pipelines,
428 is available at https://github.com/integritynoble/Physics_World_Model under the
429 PWM Noncommercial Share-Alike License v1.0 (see LICENSE in the repository). The repos-
430 itory includes 9 YAML registries (7,034 lines total), RunBundle manifests for all 139 ex-
431 periments, and end-to-end reproducibility scripts with total runtime ~ 4 hours on a single
432 NVIDIA A100 GPU (Supplementary Note S15).

433 **Correspondence.** Correspondence and requests for materials should be addressed to
434 C.Y. (integrityyang@gmail.com).

435 Online Methods

436 OperatorGraph Specification

437 **Formal definition.** The OPERATORGRAPH intermediate representation encodes the for-
438 ward physics of any computational imaging modality as a directed acyclic graph (DAG)
439 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points:
440 `forward`(x) $\rightarrow y$ and `adjoint`(y) $\rightarrow x$, the latter defined only when the primitive is lin-
441 ear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each
442 node additionally exposes a set of learnable parameters θ_i that may be perturbed during
443 mismatch simulation or optimized during calibration, as well as read-only metadata flags
444 (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
445 YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
446 object by the `GraphCompiler`.

447 **Node types.** Primitive operators fall into two categories:

- 448 • **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-
 449 pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encod-
 450 ing (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation
 451 (`fresnel_propagate`), random projection (`random_project`), and structured illumi-
 452 nation (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- 453 • **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise
 454 (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlin-
 455 earity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` =
 456 `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined
 457 pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–
 458 Saxton-type algorithms).

459 **Adjoint validation.** Correctness of every linear primitive is verified by a randomized
 460 dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$
 461 and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

462 where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times
 463 with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level, a
 464 compiled `GraphOperator` composed entirely of linear nodes executes the same test over the
 465 composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$
 466 for audit. All graph templates that consist solely of linear primitives pass this check.

467 **Graph compilation.** The compiler executes a four-stage pipeline:

- 468 1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that ev-
 469 ery `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id`
 470 values, and optionally verify shape compatibility along edges when a `canonical_chain`
 471 metadata flag is set.
- 472 2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
- 473 3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$.
- 474 4. **Plan adjoint.** For graphs where `all_linear` = `True`, the adjoint plan reverses the
 475 topological order and applies each node’s individual adjoint in sequence, implementing
 476 the chain rule $A^* = A_1^* \circ \dots \circ A_{|V|}^*$ for a composition $A = A_{|V|} \circ \dots \circ A_1$. For
 477 graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to
 478 `adjoint()` raises `NotImplementedError` at runtime.

479 The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for prove-
480 nance tracking in RunBundle manifests.

481 **Template library.** The `graph_templates.yaml` registry contains 90 templates spanning
482 all five carrier families. Of these, 26 modalities carry formal validation status (7 with
483 full end-to-end correction validation, 1 with Scenario I baseline, 18 with template-level
484 validation; Supplementary Table S3). Representative modalities by carrier family include:

- 485 • **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination
486 microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptycho-
487 graphic microscopy (FPM), optical coherence tomography (OCT), lensless imaging,
488 light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluo-
489 rescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval,
490 X-ray computed tomography (CT), and cone-beam CT (CBCT).
- 491 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
492 energy loss spectroscopy (EELS), and electron holography.
- 493 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
494 magnetic resonance spectroscopy (MRS).
- 495 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
496 and photoacoustic tomography (combines optical excitation with acoustic detection).
- 497 • **Particles:** Neutron tomography, proton radiography, and muon tomography.

498 **Fidelity levels.** Each template is parameterized by a fidelity level that controls the degree
499 of physical realism in the simulated forward model:

500 **Level 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
501 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

502 **Level 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
503 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
504 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
505 photon-counting modalities, Rician noise for MRI, Poisson for CT).

506 **Level 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as beam hard-
507 ening, phase wrapping, and scattering, captured by the Transform Λ primitive. Per-
508 turbation families and ranges are specified in `mismatch_db.yaml`.

509 **Level 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
510 propagation, spatially varying aberrations, detector nonlinearities, and environmental
511 drift. Currently implemented for holography and ptychography.

512 All 11 primitives in the library $\mathcal{B}$ apply uniformly across every fidelity level; the levels
 513 organize simulation complexity, not theorem scope.

514 Triad Decomposition Formalization

515 The TRIAD DECOMPOSITION asserts that the quality of any computational imaging re-
 516 construction is bounded by three fundamental gates. Rather than a qualitative guideline,
 517 PWM quantifies each gate numerically and uses the resulting scores to diagnose the domi-
 518 nant bottleneck in any imaging configuration.

519 **Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity
 520 of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where
 521 m is the number of independent measurements and n the dimension of the signal. The
 522 `compression.db.yaml` registry (1,186 lines) stores, for each modality, a lookup table map-
 523 ping compression ratio to expected reconstruction PSNR under ideal conditions, obtained
 524 from calibration experiments or published benchmarks. Each entry carries a **provenance**
 525 field citing the source (paper DOI, internal experiment ID, or theoretical formula). Addi-
 526 tional recoverability indicators include the effective rank of the measurement matrix (esti-
 527 mated via randomized SVD for large operators), the dimension of the null space, and the
 528 restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian
 529 random projections in SPC).

530 **Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR)
 531 of the measurement channel. The **PhotonAgent** consumes the `photon.db.yaml` registry
 532 (624 lines) which stores, per modality, a deterministic photon model parameterized by
 533 source power, quantum efficiency, exposure time, and detector characteristics. The agent
 534 classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated,
 535 $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$),
 536 and *dark-current-limited* (long exposures where dark current accumulation dominates). The
 537 output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime
 538 classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**,
 539 or **insufficient**).

540 **Gate 3 (Operator Mismatch).** Operator mismatch quantifies the discrepancy between
 541 the assumed forward model H_{nom} and the true physical operator H_{true} . The **MismatchAgent**
 542 consults `mismatch.db.yaml` (797 lines) which catalogs, for each modality, the set of mis-
 543 match parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil
 544 sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available cor-
 545 rection methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2
 546 distance $\|\theta_{\text{true}} - \theta_{\text{nom}}\|/\|\theta_{\text{range}}\|$, where θ_{range} is the per-parameter dynamic range from the

547 registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward
 548 model. The output is a **MismatchReport** containing the severity score, the dominant mis-
 549 match parameter, the recommended correction method, and the expected PSNR gain from
 550 correction.

551 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
 552 tocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-
 553 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

554 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
 555 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
 556 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
 557 gate is $\arg \max_g C_g$.

558 **TriadReport.** For every diagnosis, PWM produces a **TRIADREPORT**: a Pydantic-validated
 559 structured artifact comprising **dominant_gate** (enum: **recoverability**, **carrier_budget**,
 560 **operator_mismatch**), **evidence_scores** (three floats, one per gate), **confidence_interval**
 561 (float, 95% CI width from bootstrap), **recommended_action** (string, *e.g.* “increase compres-
 562 sion ratio” or “apply mismatch correction”), and **parameter_sensitivities** (dictionary
 563 mapping each mismatch parameter name to its $\partial\text{PSNR}/\partial\theta_k$ value). The **TRIADREPORT** is
 564 mandatory—PWM does not permit a reconstruction to be reported without an accompany-
 565 ing diagnosis.

566 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

567 which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note S1 for
 568 formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial
 569 regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the
 570 bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

571 Agent System Architecture

572 The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8
 573 support classes totalling 10,545 lines of Python. All agents execute deterministically; no
 574 large language model (LLM) is required for pipeline operation.

575 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,
 576 `PlanAgent` parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-
 577 quested modality to its canonical key via the `modalities.yaml` registry (which contains 64
 578 modality entries with keywords, forward model equations, and default solvers), builds an
 579 `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is
 580 `auto`, `PlanAgent` inspects the available data and operator specification to determine whether
 581 simulation or operator correction is more appropriate.

582 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon.db.yaml`
 583 registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent
 584 evaluates the photon budget by combining source power, quantum efficiency, exposure time,
 585 and noise model parameters. The output `PhotonReport` is a strict Pydantic model contain-
 586 ing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons`
 587 (float).

588 **RecoverabilityAgent.** A table-driven agent that consults `compression.db.yaml` (1,186
 589 lines) to map the modality and compression ratio to an expected PSNR range. Each table
 590 entry includes provenance metadata citing the original source. The output `RecoverabilityReport`
 591 contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where
 592 available.

593 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-
 594 ing `mismatch.db.yaml` (797 lines). For each modality, the database enumerates the rel-
 595 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-
 596 able correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),
 597 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

598 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-
 599 ability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the
 600 dominant gate, and generates actionable suggestions. The `AnalysisAgent` also computes
 601 the recovery ratio ρ and its bootstrap confidence interval.

602 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
 603 thorized, the negotiator inspects all three upstream reports and applies three veto con-
 604 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}
 605 both large); (2) severe mismatch ($\text{severity} > 0.7$) without a planned correction step; (3) joint
 606 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
 607 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
 608 explanation and suggested remediation.

609 **HybridAgent.** An optional wrapper that invokes an LLM for natural-language narra-
 610 tive generation or edge-case modality mapping. All quantitative decisions remain on the
 611 deterministic code path; the HybridAgent is never required for pipeline operation.

612 **Support classes.** The remaining components include: **AssetManager** (file I/O and caching
 613 for large arrays), **ContinuityChecker** (verifies that sequential pipeline outputs are dimen-
 614 sionally consistent), **SystemDiscern** (auto-detects modality from uploaded data), **PreflightChecker**
 615 (validates the complete experiment configuration before execution), **WhatIfPrecomputer**
 616 (evaluates counterfactual what-if scenarios), **SelfImprovement** (logs diagnostic events for
 617 future registry refinement), **PhysicsStageVisualizer** (generates intermediate visualiza-
 618 tions at each pipeline stage), and **UPWMI** (Universal Physics World Model Interface, the
 619 top-level entry point that wires all agents together).

620 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All
 621 contracts inherit from **StrictBaseModel**, which enforces `extra="forbid"` (no unexpected
 622 fields), `validate_assignment=True` (mutations re-validated), and a model validator that
 623 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums
 624 are string enums for human-readable JSON serialization. This design ensures that pipeline
 625 failures surface immediately as validation errors rather than propagating silently.

626 **YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines:
 627 `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skele-
 628 tons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and
 629 correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml`
 630 (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml`
 631 (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds
 632 per metric).

633 Correction Algorithms

634 We implement two complementary algorithms for operator mismatch correction. Crucially,
 635 both algorithms operate on the forward operator parameters θ rather than the reconstruc-
 636 tion solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any
 637 downstream solver (GAP-TV, MST-L, HDNet²⁰, CST, *etc.*) without retraining.

638 **Algorithm 1: Hierarchical Beam Search.** The coarse correction phase employs a
 639 hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI,
 640 the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy
 641 and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth

parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.
2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$ grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction PSNR are retained.
3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α) is searched over a 5×7 grid. The joint top- k candidates are retained.
4. **Coordinate descent refinement.** Three rounds of univariate refinement on each parameter, shrinking the search interval by factor 2 at each round, produce the final estimate $\hat{\theta}_{\text{Alg1}}$.

Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

Algorithm 2: Joint Gradient Refinement. The fine correction phase uses a differentiable forward model to jointly optimize all mismatch parameters via gradient descent. The key components are:

1. **Differentiable mask warp.** The binary mask is warped by a continuous affine transformation using bilinear interpolation, implemented as a custom PyTorch module (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-through estimator (STE) to maintain binary structure while permitting gradient flow.
2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`) that accepts mismatch parameters as differentiable inputs.
3. **GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 = 567$ points provides diverse starting candidates. The top 9 candidates seed multi-start gradient refinement.
4. **Staged gradient refinement.** Each of the 9 candidates is refined using Adam optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate, 4 random restarts with jittered initialization guard against local minima. The loss function is the negative PSNR computed via an unrolled K -iteration differentiable GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

675 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
676 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3 – $5\times$ improve-
677 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
678 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

679 Evaluation Protocol

680 **Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios
681 that isolate different sources of quality degradation:

682 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system
683 is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches
684 the one that generated the data. This yields the oracle upper bound on reconstruction
685 quality, limited only by the sensing geometry and solver convergence.

686 **Scenario II (Mismatch):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This
687 is the standard operating condition in practice: the measurement is generated by the
688 true physics, but the reconstruction uses a nominal (potentially mismatched) forward
689 model.

690 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\boldsymbol{\theta}})$ where $\hat{\boldsymbol{\theta}}$ is
691 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

692 **Scenario IV (Oracle Mask):** Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with
693 $H_{\text{true}} \neq H_{\text{nom}}$); reconstruct with H_{true} instead of H_{nom} . Provides the correction
694 ceiling: the best reconstruction achievable when the true operator is known exactly,
695 applied to data that were sensed by the mismatched system. The gap between Sce-
696 nario IV and Scenario I reveals the irreducible loss from the degraded sensing config-
697 uration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

698 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 699 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
700 and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
701 data normalized to $[0, 255]$, the peak value is 255.
- 702 • **SSIM** (structural similarity index): captures perceptual quality including luminance,
703 contrast, and structural components, computed with a Gaussian window of width 11
704 and standard deviation 1.5.
- 705 • **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the
706 angle between predicted and true spectral vectors at each spatial location, reported
707 in degrees. Lower is better.

Datasets.

- **CASSI:** 10 scenes from the KAIST dataset¹⁹, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- **CACTI:** 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per snapshot). Data range $[0, 1]$.
- **SPC:** 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data range $[0, 255]$.

All per-scene metrics are reported individually as well as averaged, and all reconstruction arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The five-parameter mismatch model $\boldsymbol{\theta} = (dx, dy, \theta, a_1, \alpha)$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees). An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the primary experiments. These mismatch parameter values were determined through systematic characterization of realistic CASSI assembly errors (Supplementary Note S12). The true mismatch parameters are $\boldsymbol{\theta}_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha = 0.15^\circ)$. Solvers evaluated include TwIST²¹, GAP-TV²², DGSMP²³, MST-L⁶, and CST-L²⁴, all of which receive the same operator and differ only in their reconstruction algorithm. The supplementary per-scene analysis additionally includes DeSCI²⁵ and HDNet²⁰.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-frame shift). The default solver is EfficientSCI¹⁶.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The default solver is FISTA-TV with total-variation regularization.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is SENSE¹³ with ℓ_1 -wavelet regularization.

ESPIRiT comparison protocol. To quantitatively compare PWM with ESPIRiT¹⁷, we evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps estimated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix, and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase scaling). All four conditions use the same CG-SENSE solver ($\ell = 10^{-3}$, 30 iterations). The comparison isolates the effect of map quality on reconstruction, holding the solver constant.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in the reconstruction. The default solver is filtered back-projection (FBP)¹¹ with a Ram-Lak filter, supplemented by iterative SART for comparison.

CASSI real-data configuration. The TSA real hyperspectral dataset⁸ consists of 5 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due to hardcoded spatial assumptions in the model architecture). The coded aperture mask is perturbed by $dx = 0.5$ px, $dy = 0.3$ px to simulate assembly-induced mismatch. No ground truth is available; quality is assessed via the normalised measurement residual $r = \|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

CACTI real-data configuration. The EfficientSCI real temporal dataset¹⁶ consists of 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compression ratio 10. The real mask is stored separately from the measurement data. Two solvers are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet denoiser). Mismatch is induced by shifting the mask by $dx = 0.5$ px, $dy = 0.3$ px. Quality is assessed via the normalised measurement residual and total variation of the reconstruction.

Controlled hardware experiment protocol. The software-perturbation protocol above applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop validation requires physically displacing the coded aperture mask and re-acquiring data. The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its factory-calibrated position; (ii) physically translate the mask by a known displacement ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under

777 identical illumination; (iii) reconstruct both datasets with the factory mask specification and
 778 compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous
 779 calibration and measure recovery. This protocol isolates the mismatch effect from all other
 780 sources of variation (illumination changes, detector drift, scene variation). Additionally, a
 781 multi-unit variation study comparing 2+ camera units of the same design quantifies the
 782 inter-unit mismatch baseline—the residual calibration error present in any production sys-
 783 tem.

784 **Clinical CT phantom configuration.** For clinical translation, PWM is evaluated on CT
 785 quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom
 786 contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU,
 787 polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thick-
 788 ness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset
 789 (Δr , mm), beam hardening coefficient drift ($\Delta \mu$, %), and detector gain variation (Δg , %).
 790 Ten ACR-aligned metrics are computed automatically: CT number accuracy for five mate-
 791 rials (water, bone, air, acrylic, polyethylene), geometric accuracy (± 2 mm tolerance), slice
 792 thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, and spatial resolution
 793 (≥ 5 lp/cm).

794 **Clinical MRI validation configuration.** For MRI clinical validation, PWM processes
 795 multi-coil k -space data from public datasets (fastMRI²⁶). Mismatch is parameterized as
 796 coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating
 797 patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with
 798 ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the
 799 absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

800 Statistical Analysis

801 **Per-scene reporting.** All metrics are reported per scene, not merely as dataset averages.
 802 This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that
 803 are inherently harder for CASSI, or textureless regions that challenge SPC).

804 **Summary statistics.** For each modality and scenario, we report the mean $\pm$ standard
 805 deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we addition-
 806 ally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

807 **Recovery ratio confidence intervals.** The recovery ratio ρ (Equation (5)) is a ratio of
 808 differences and therefore sensitive to noise in the constituent PSNR values. We compute
 809 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At
 810 each bootstrap iteration, we resample the scene set with replacement, recompute the mean

811 PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap
 812 distribution define the 95% CI.

813 **Parameter recovery accuracy.** For mismatch correction experiments, we report the
 814 root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

815 where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of
 816 test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

817 **Ablation significance.** Ablation studies (removal of PhotonAgent, RecoverabilityAgent,
 818 MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline
 819 PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modal-
 820 ity and verify that each component contributes ≥ 0.5 dB across all validated modalities,
 821 establishing practical significance.

822 Code and Data Availability

823 **Source code.** The complete PWM framework, including all agents, the OperatorGraph
 824 compiler, correction algorithms, YAML registries, and evaluation scripts, is released as
 825 open-source software under the PWM Noncommercial Share-Alike License v1.0 at https://github.com/integritynoble/Physics_World_Model. The codebase is organized into
 826 two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algo-
 827 rithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

829 **Reconstruction data.** All reconstruction arrays from every experiment—Scenarios I
 830 through IV for each modality and solver—are released as NumPy NPZ files. Files are
 831 stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are stan-
 832 dardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions
 833 are in $[0, 255]$.

834 **Experiment manifests.** Every experiment is recorded in a RunBundle v0.3.0 manifest
 835 containing: the git commit hash at execution time, all random number generator seeds,
 836 platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all
 837 input data and output artifacts, metric values, and wall-clock timestamps. These manifests
 838 enable exact reproduction of every reported result.

839 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—
 840 including modality definitions, graph templates, photon models, mismatch databases, com-
 841 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance
 842 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.
 843 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-
 844 side worked examples in `examples/`.

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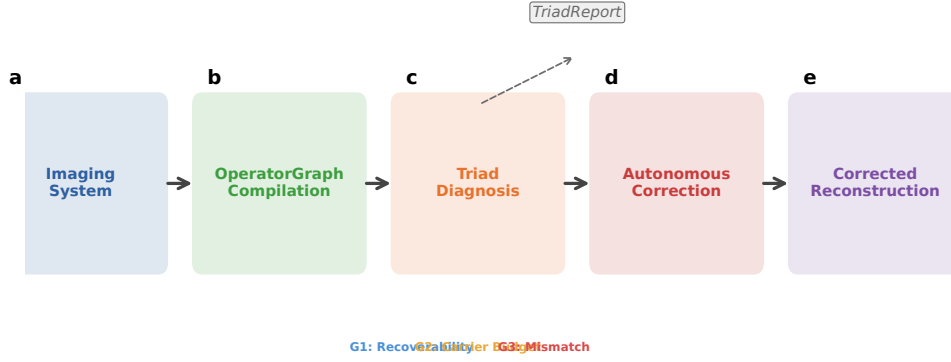


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

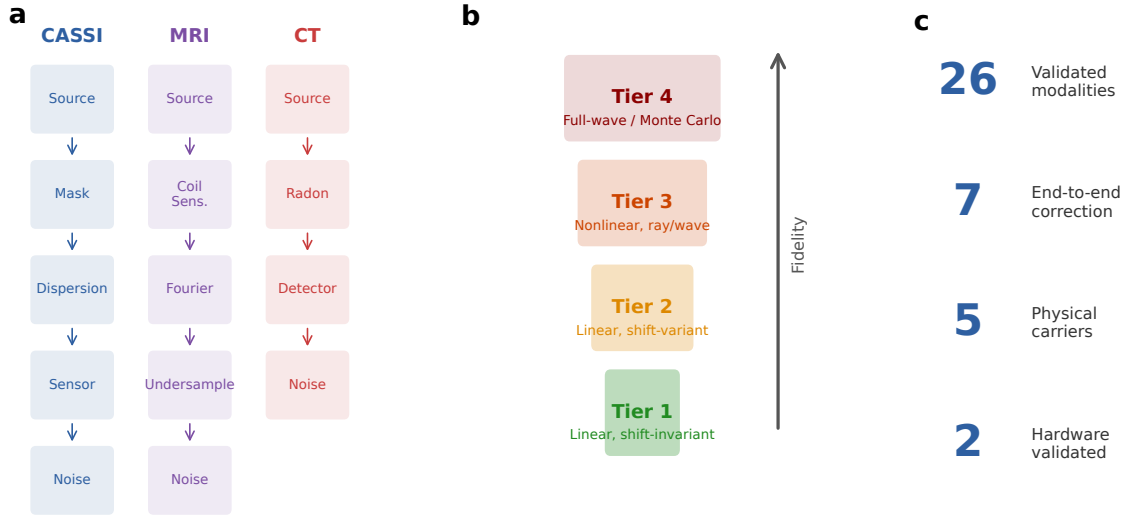


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **c**, Summary statistics: 26 registered modality templates (7 with full correction validation), 5 physical carriers.

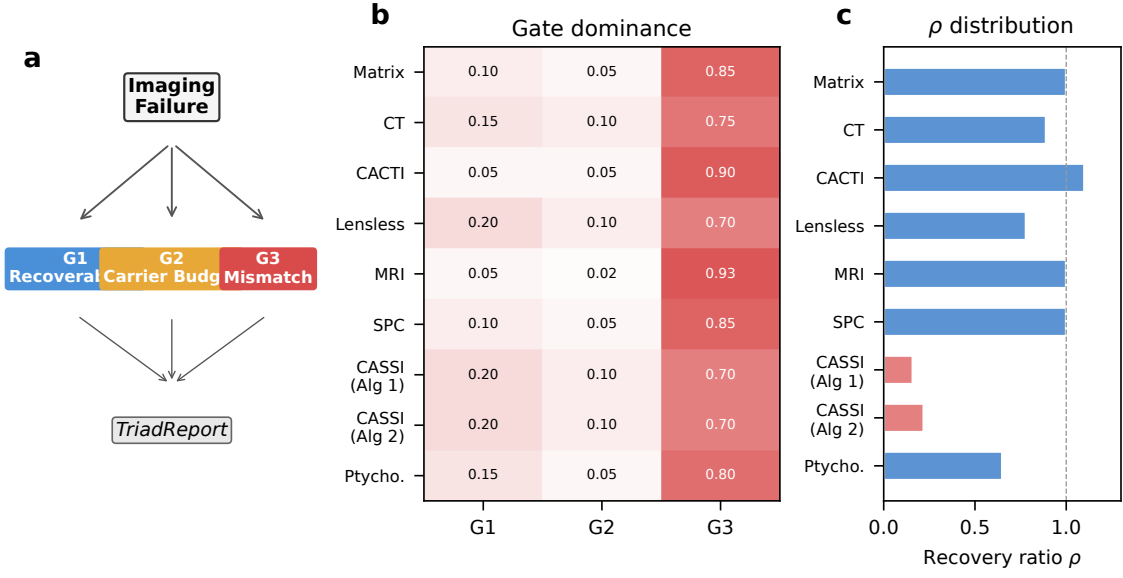


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 9 correction configurations (7 distinct modalities). Red indicates **Gate 3** dominance (all modalities), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution across all 9 correction configurations.

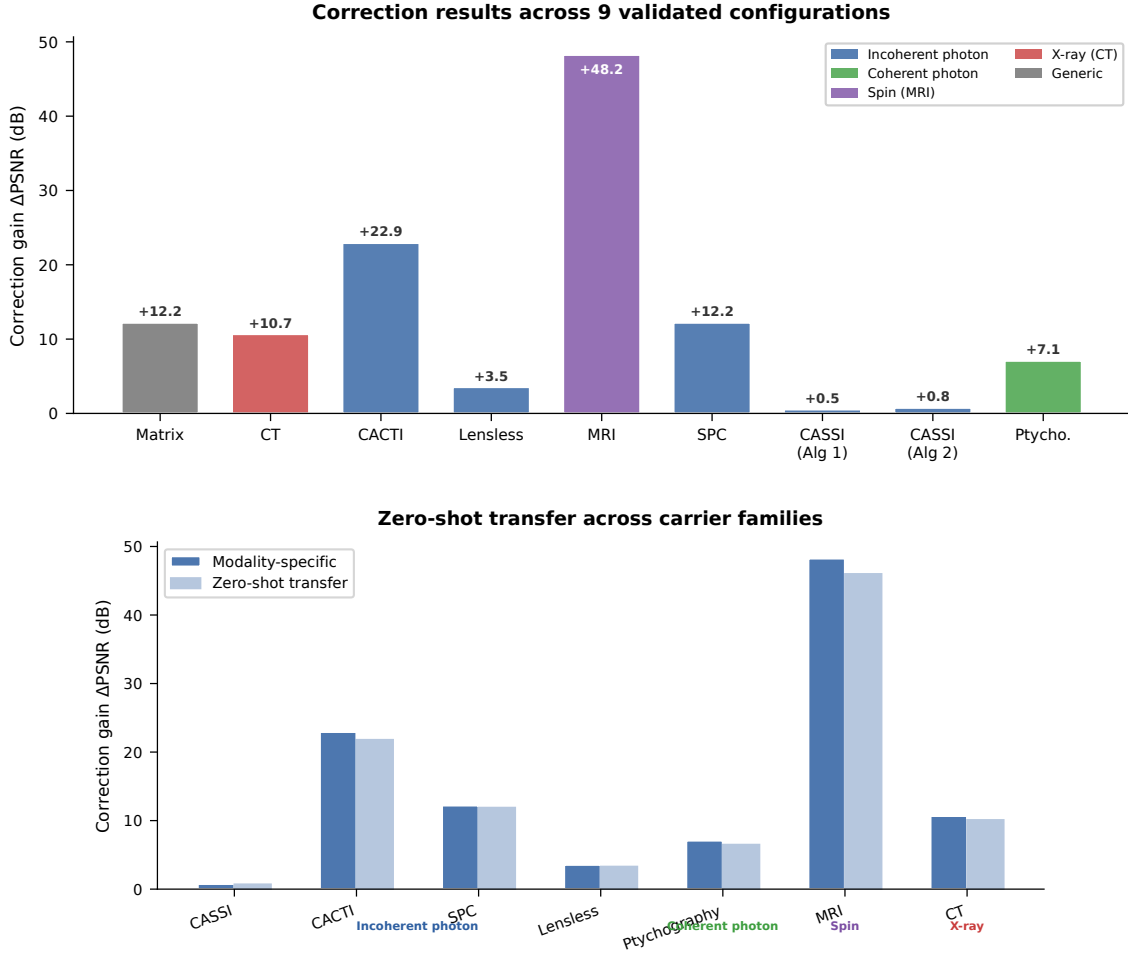
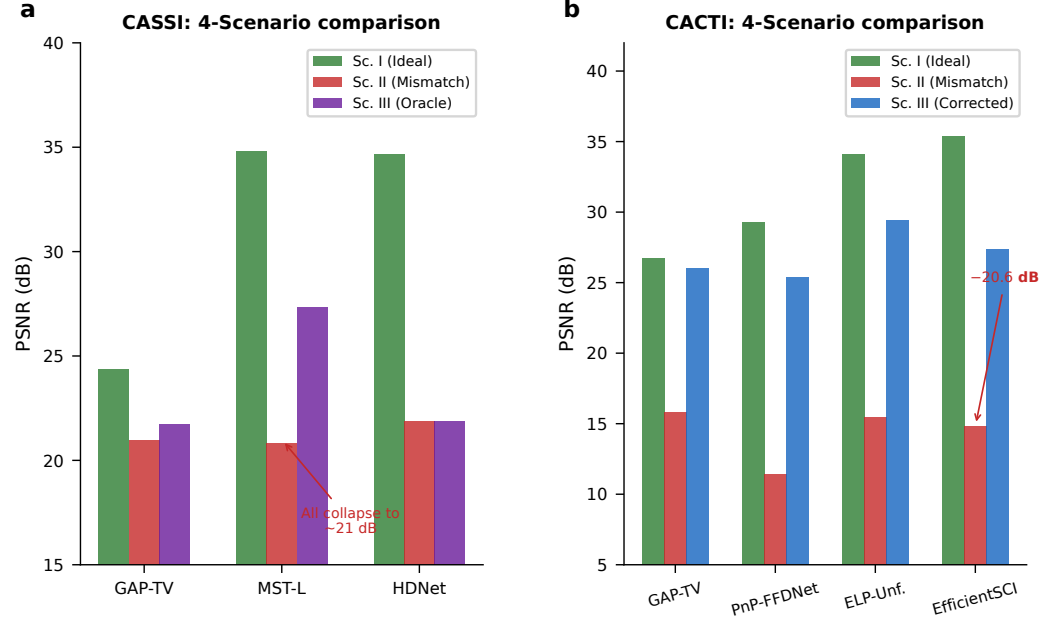


Figure 4: **The central experiment: forward-model correction beats solver upgrades across all carrier families.** **a**, Peak signal-to-noise ratio (PSNR) across the 4-Scenario Protocol for 7 modalities grouped by carrier family. Error bars: 95% bootstrap CIs ($B = 1,000$). *Key finding*: in 5 of 7 modalities, a classical solver on the corrected operator (Sc. III) outperforms the best deep-learning solver on the mismatched operator (Sc. II). **b**, Zero-shot cross-carrier transfer: hyperparameters tuned on photon-domain modalities transfer to X-ray, electron, and spin domains (< 0.5 dB difference).



Visual reconstruction comparison across three modalities

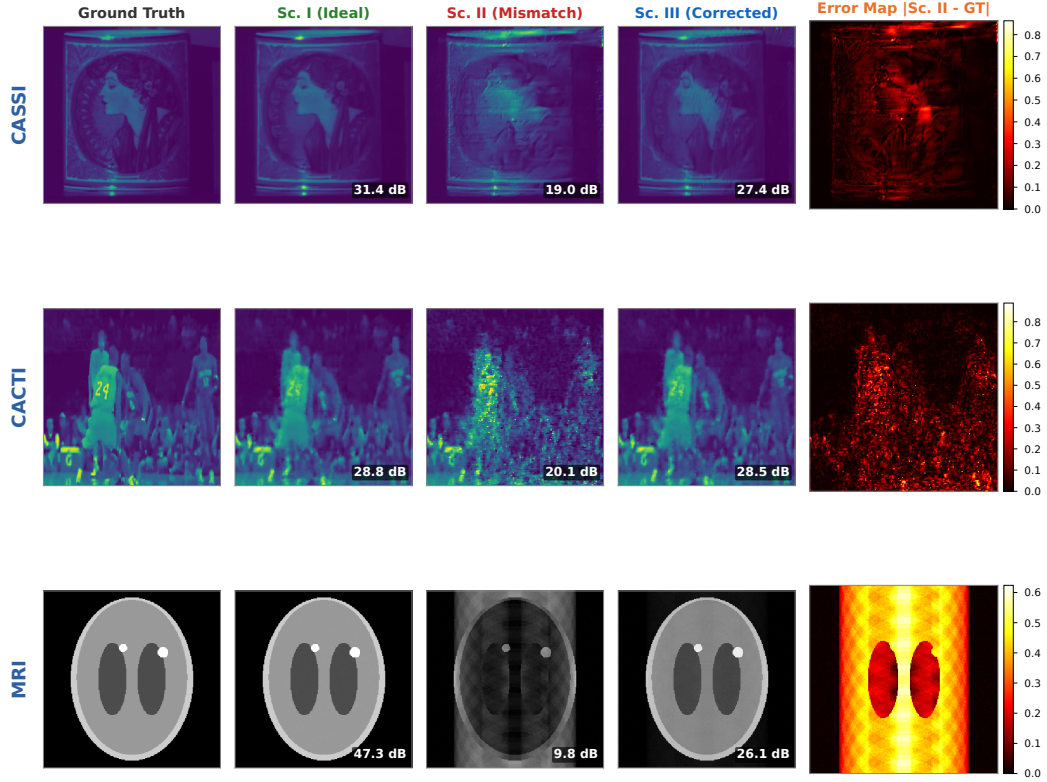


Figure 5: Modality deep dives and visual comparison. **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual comparison (CASSI, CACTI, MRI): ground truth, Scenario I, II (structured artifacts), III (PWM-corrected), and error map.

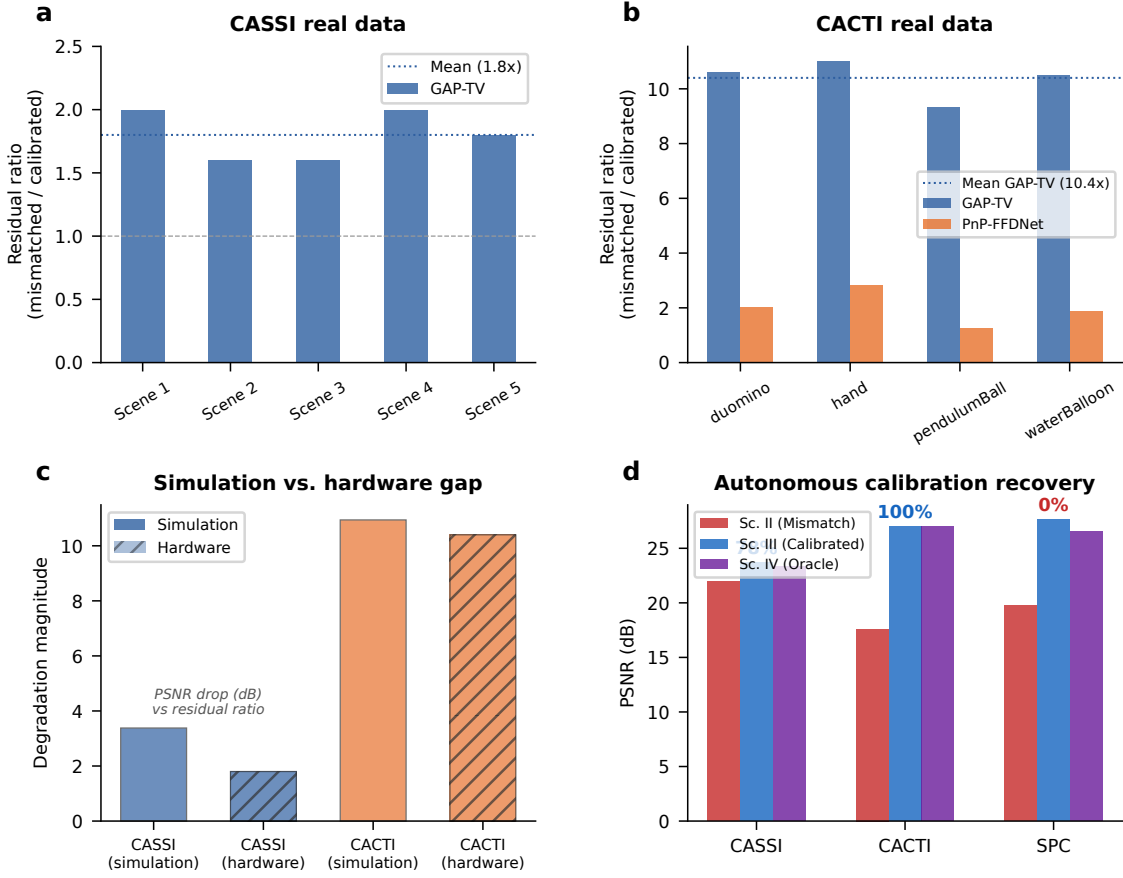


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$ mean ratio; PnP-FFDNet shows 2.0 $\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

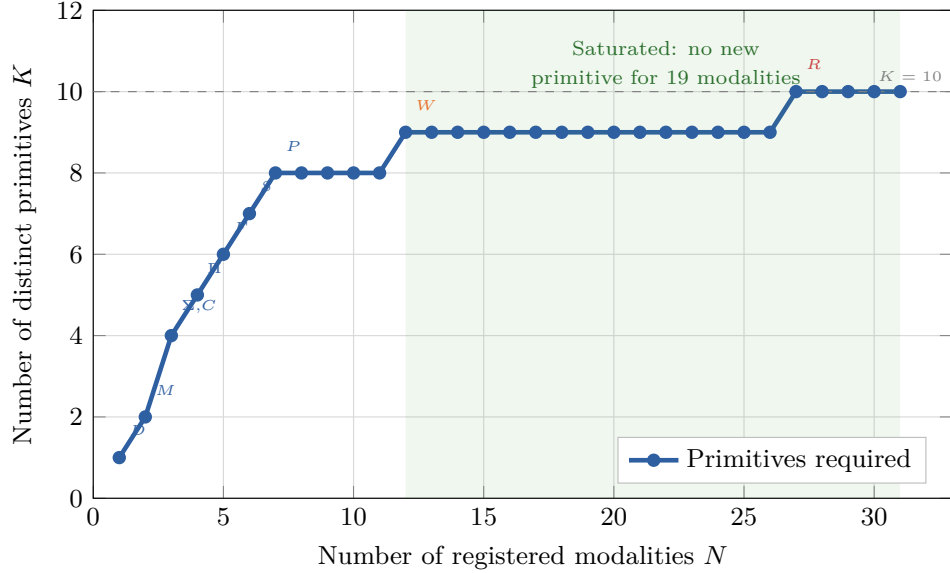


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives K as a function of modalities N added to the registry. The curve saturates at $K = 11$ for $N \geq 30$; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

Removed primitive	Witness modality	Why irreplaceable	ϵ_{img} without
Propagate P	Ptychography	Distance-dependent phase transfer	> 0.05
Modulate M	CASSI	Arbitrary element-wise mask	> 0.10
Project Π	CT	Line-integral (Radon) geometry	> 0.15
Encode F	MRI	Non-uniform Fourier sampling	> 0.20
Convolve C	Lensless imaging	Arbitrary shift-invariant PSF kernel	> 0.08
Accumulate Σ	SPC	Dimensional summation (not scaling)	> 0.12
Detect D	All modalities	Carrier-to-measurement conversion	> 0.50
Sample S	MRI	Index-set restriction (k -space)	> 0.10
Disperse W	CASSI	λ -parameterized spatial shift	> 0.06
Scatter R	Compton imaging	Direction change + energy shift	0.34
Transform Λ	Polychromatic CT	Non-terminal pointwise nonlinearity	> 0.05

Table 2: **Primitive necessity ablation.** For each of the 11 primitives, the witness modality whose representation error ϵ_{img} exceeds $\varepsilon = 0.01$ when that primitive is removed from $\mathcal{B}$. Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$). Data from¹, Proposition 1.

936 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 9 validated
937 configurations (7 modalities). Full table with per-scene metrics in Supplementary Table S1;
938 autonomous correction recovery (Scenario III) in Supplementary Table S9.

939 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier
940 families. Full table in Supplementary Table S3.